

Xiaochang Li

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EDUCATION

Carnegie Mellon University

Master of Science in Mobile and IoT Engineering

Pittsburgh, PA

August 2025 - May 2027

- GPA: 4.0/4.0
- Coursework: Introduction to Computer Systems, Cloud Infrastructure and Services, Database Systems

Fudan University

Bachelor of Science in Computer Science

Shanghai, China

September 2020 - August 2024

- GPA: 90/100
- Coursework: Data Structures & Algorithms, Computer Systems, Computer Networks, Parallel Distributed Computing
- Honors & Reward: Scholarship of Fudan University, Fudan Outstanding Student

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, TypeScript, Terraform

Frameworks & Libraries: Spring Boot, PyTorch, Pytest, Google Test, DPDK, MapReduce, Spark, Yarn

Tools & Platforms: Git, Linux, Docker, MySQL, Kubernetes, Kafka, AWS (EC2, S3, Lambda, ECS), GCP(GKE, Dataproc, App Engine, VPC), GitLab CI/CD, Jenkins, SonarQube

Concepts: Data Structures & Algorithms, Software Testing, Pub/Sub, IPC, Distributed Systems, OOD, Microservices, IaC

WORK EXPERIENCE

Tiktok

Shanghai, China

SOFTWARE ENGINEERING INTERN

December 2024 - April 2025

- Identified bottlenecks in test data/case management; developed CI/CD workflows automating build, test (Pytest), and release, cut end-to-end duration by 20% and saving 250+ annual hours
- Spearheaded LLM-based DOM analysis framework generating 85% of functional test cases; authored 200+ backend Pytest suites, collectively increasing overall code coverage from 70% to 80%
- Collaborated with Product/UI to co-author 50+ API specs for Stream Resource features, ensuring clear contracts and cutting down integration friction

Xiaoyong Private Equity Funds

Shanghai, China

SOFTWARE ENGINEER INTERN

June 2023 - September 2023

- Optimized IPC for high-frequency market data feeds by replacing UDS with DPDK/libmemif zero-copy channels, attaining 23% lower latency (2.4 μ s $\rightarrow$ 1.84 μ s in bulk data transfers under localhost benchmark)
- Refactored messaging system from scratch to custom C++ pub-sub model, decoupling producers/consumers and enhancing flexibility/scalability
- Authored 80+ Google Test suites for IPC libraries, achieving 85% coverage and bolstering robustness
- Automated partner report processing from Excel in email to MySQL with Pandas, eliminating 30 min/report manual effort

KEY PROJECTS

Automated CI/CD Pipeline with Quality Gates | Terraform, GKE, Jenkins, SonarQube, Dataproc, Helm

- Provisioned end-to-end CI/CD infrastructure on GCP using Terraform IaC: GKE cluster, Jenkins/SonarQube (Helm charts), and 3-node Dataproc cluster, reducing deployment time from 120min to 15min
- Built GitHub web-hook triggered pipeline performing incremental SonarQube analysis on changed files only, enforcing quality gates to conditionally trigger Hadoop MapReduce jobs
- Designed Kubernetes-native Jenkins agents with GCloud SDK containers for seamless GCP authentication, enabling automated Dataproc job submission Yarn job management

High-Performance Dynamic Memory Allocator | C, Systems Programming

- Implemented segregated free lists with 12 size classes and mini-block optimization for 8-16B allocations, achieving 74% memory utilization and 9,662 Kops/sec throughput (100/100 performance score)
- Eliminated footers for allocated blocks by encoding previous allocated status in header bits, reducing metadata overhead by 50% while maintaining O(1) coalescing through bi-directional list traversal
- Optimized for SPEC benchmarks through first-fit policy and immediate coalescing, outperforming baseline (7,242 Kops/sec) by 33% in all workloads

Scalable Sentiment Analysis App | Kubernetes, Docker, Spring Boot, Python, GKE

- Deployed 3-tier sentiment analysis app on GKE: React frontend, Spring Boot middleware, Python ML backend, accomplishing decoupled architecture with independent scaling per service
- Integrated Kafka between WebApp and Logic layers for async message processing